

ECO491 PSET HW5

Written Solutions

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Question 1: Hedging Interest Rate Risk

The term structure is given by

$$R(T) = \beta_1 + \beta_2 T + \beta_3 T^2, \quad (\beta_1, \beta_2, \beta_3) = (0.04, 0.012, -0.0015), \quad D(T) = e^{-R(T)T}.$$

The 8-year 6% semiannual bond (face 100) pays cash flows 3 at $T_k = 0.5k$, $k = 1, \dots, 15$, and 103 at $T_{16} = 8$.

1.1 Bond price

The price follows from discounting each payment: $B_t = \sum_{k=1}^{16} \text{CF}_k e^{-R(T_k)T_k}$. Representative terms are:

k	T_k	CF_k	$R(T_k)$	$\text{CF}_k e^{-R(T_k)T_k}$
1	0.5	3	0.0450	2.934
2	1.0	3	0.0505	2.854
3	1.5	3	0.0548	2.771
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
14	7.0	3	0.0820	2.168
15	7.5	3	0.0829	2.134
16	8.0	103	0.0832	84.998

Summing all discounted flows gives $B_t \approx 110.858$

1.2 Level and curvature factor durations

Write the curve as $R(T) = \beta_1 f_{\text{lev}}(T) + \beta_2 f_{\text{slope}}(T) + \beta_3 f_{\text{curv}}(T)$, $f_{\text{lev}}(T) = 1$, $f_{\text{curv}}(T) = T^2$. For a generic bond:

$$P = \sum_{k=1}^n \text{CF}_k e^{-R(T_k)T_k},$$

and the factor duration with respect to factor j is:

$$D_j = -\frac{1}{P} \frac{\partial P}{\partial \beta_j} = \frac{1}{P} \sum_{k=1}^n \text{CF}_k T_k f_j(T_k) e^{-R(T_k)T_k}.$$

For the 8-year bond:

$$D_{\text{lev},B} = \frac{1}{B_t} \sum_{k=1}^{16} \text{CF}_k T_k e^{-R(T_k)T_k} \approx \boxed{6.619}, \quad D_{\text{curv},B} = \frac{1}{B_t} \sum_{k=1}^{16} \text{CF}_k T_k^3 e^{-R(T_k)T_k} \approx \boxed{380.495}$$

A zero maturing at T has price $Z_t(T) = 100e^{-R(T)T}$ and factor durations $D_{\text{lev}}^z(T) = T, D_{\text{curv}}^z(T) = T^3$. For the 1-year and 7-year zeros:

	$Z_t(T)$	$D_{\text{lev}}^z(T)$	$D_{\text{curv}}^z(T)$
$T = 1$	95.075	1	1
$T = 7$	70.223	7	343

1.3 Factor-duration hedge

The hedged portfolio: $P_t = B_t + k_1 Z_t(1) + k_7 Z_t(7)$ has factor-dollar durations:

$$FD_{\text{lev}} = B_t D_{\text{lev},B} + k_1 Z_t(1) D_{\text{lev}}^z(1) + k_7 Z_t(7) D_{\text{lev}}^z(7),$$

$$FD_{\text{curv}} = B_t D_{\text{curv},B} + k_1 Z_t(1) D_{\text{curv}}^z(1) + k_7 Z_t(7) D_{\text{curv}}^z(7).$$

Setting both exposures to zero yields

$$\begin{cases} B_t D_{\text{lev},B} + k_1 Z_t(1) + 7k_7 Z_t(7) = 0, \\ B_t D_{\text{curv},B} + k_1 Z_t(1) + 343k_7 Z_t(7) = 0. \end{cases}$$

Using $B_t = 110.858$, $D_{\text{lev},B} = 6.619$, $D_{\text{curv},B} = 380.495$, $Z_t(1) = 95.075$, $Z_t(7) = 70.223$, the solution is $k_1 \approx 1.364, k_7 \approx -1.757$ such that I would long 1.364 units of the 1-Year bond and short 1.757 units of the 7-year bond.

1.4 Hedge effectiveness under level+curvature shifts

Consider ten scenarios with

$$\beta_1^{(i)} = 0.04 + i \frac{0.025}{10}, \quad \beta_3^{(i)} = -0.0015 + i \frac{0.002}{10}, \quad \beta_2^{(i)} = 0.012.$$

Let $B_{t+1}^{(i)}$ and $Z_t^{(i)}(1), Z_t^{(i)}(7)$ be the revalued securities, and define

$$P_{t+1}^{(i)} = B_{t+1}^{(i)} + k_1 Z_t^{(i)}(1) + k_7 Z_t^{(i)}(7), \quad \Delta B^{(i)} = B_{t+1}^{(i)} - B_t, \quad \Delta P^{(i)} = P_{t+1}^{(i)} - P_t.$$

With $P_t \approx 117.191$, numerical results are

i	$B_{t+1}^{(i)}$	$P_{t+1}^{(i)}$	$\Delta B^{(i)}$	$\Delta P^{(i)}$
1	101.159	117.318	-9.699	0.128
2	92.516	117.664	-18.342	0.473
3	84.811	118.177	-26.048	0.986
4	77.937	118.816	-32.921	1.626
5	71.802	119.546	-39.057	2.356
6	66.322	120.338	-44.537	3.147
7	61.424	121.167	-49.435	3.976
8	57.043	122.012	-53.816	4.822
9	53.121	122.859	-57.738	5.668
10	49.607	123.693	-61.251	6.502

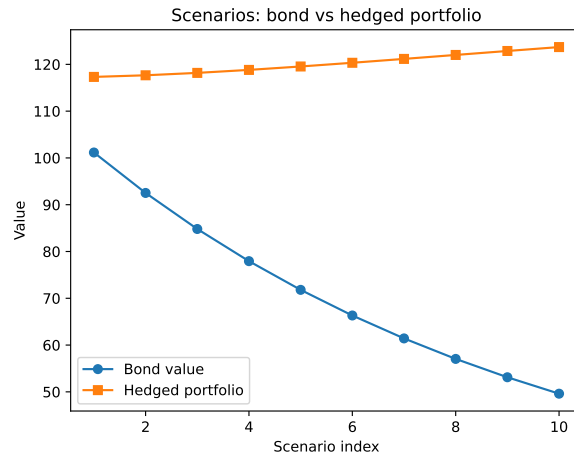


Figure 1.0.1: Historical Simulation VaR Bond

The unhedged bond declines sharply as level and curvature rise, while the hedged portfolio exhibits much smaller changes. The hedge removes first-order exposure to level and curvature; the residual variation reflects higher-order effects.

Question 2: Bond convexity

A generic bond with cash flows $\{(CF_k, T_k)\}_{k=1}^n$ has price $P = \sum_{k=1}^n CF_k e^{-R(T_k)T_k}$. A parallel shift changes $R(T_k) \rightarrow R(T_k) + dR$ for all k and

$$\frac{\partial}{\partial R} e^{-RT_k} = -T_k e^{-RT_k}, \quad \frac{\partial^2}{\partial R^2} e^{-RT_k} = T_k^2 e^{-RT_k},$$

giving:

$$\frac{\partial P}{\partial R} = \sum_{k=1}^n -CF_k T_k e^{-R(T_k)T_k}, \quad \frac{\partial^2 P}{\partial R^2} = \sum_{k=1}^n CF_k T_k^2 e^{-R(T_k)T_k}.$$

Traditional duration and convexity:

$$D = \frac{1}{P} \sum_{k=1}^n CF_k T_k e^{-R(T_k)T_k}, \quad C = \frac{1}{P} \sum_{k=1}^n CF_k T_k^2 e^{-R(T_k)T_k}.$$

For the 8-year bond:

k	T_k	T_k^2	$CF_k e^{-R(T_k)T_k}$
1	0.5	0.25	2.934
8	4.0	16.00	2.426
12	6.0	36.00	2.281
16	8.0	64.00	84.998

$$D = \frac{1}{110.858} \sum_{k=1}^{16} CF_k T_k e^{-R(T_k)T_k}$$

$$C = \frac{1}{110.858} \sum_{k=1}^{16} CF_k T_k^2 e^{-R(T_k)T_k}$$

Under the polynomial term structure:

$$R(T) = \beta_1 \cdot 1 + \beta_2 T + \beta_3 T^2, \quad f_{\text{lev}}(T) = 1, \quad f_{\text{slope}}(T) = T, \quad f_{\text{curv}}(T) = T^2,$$

while the price remains $P = \sum_{k=1}^n CF_k e^{-R(T_k)T_k}$.

Differentiating with respect to factor β_j :

$$\frac{\partial P}{\partial \beta_j} = \sum_{k=1}^n \text{CF}_k(-T_k f_j(T_k)) e^{-R(T_k)T_k}, \quad \frac{\partial^2 P}{\partial \beta_j^2} = \sum_{k=1}^n \text{CF}_k(T_k f_j(T_k))^2 e^{-R(T_k)T_k}.$$

Then factor convexity is $C_j = \frac{1}{P} \sum_{k=1}^n (T_k f_j(T_k))^2 \text{CF}_k e^{-R(T_k)T_k}$.
After, one applies the three factor loadings:

Factor	Convexity Expression
Level	$C_{\text{lev}} = \frac{1}{P} \sum C F_k T_k^2 e^{-R(T_k)T_k}$
Slope	$C_{\text{slope}} = \frac{1}{P} \sum C F_k T_k^4 e^{-R(T_k)T_k}$
Curvature	$C_{\text{curv}} = \frac{1}{P} \sum C F_k T_k^6 e^{-R(T_k)T_k}$

Question 3: Black-Scholes option pricing via risk-neutral valuation

Let the underlying price at time t be S_t and maturity be T and not that the log-return $x^* = \ln\left(\frac{S_{t+T}}{S_t}\right)$ is normal such that: $x^* \sim \mathcal{N}(\mu, \sigma_x^2)$, $\mu = T\left(r_f - \frac{1}{2}\sigma^2\right)$, $\sigma_x^2 = T\sigma^2$.
The call price with strike X is:

$$c_t = e^{-r_f T} \mathbb{E}^{\mathbb{Q}}[(S_{t+T} - X)^+] = e^{-r_f T} \int_{-\infty}^{\infty} \max(S_t e^{x^*} - X, 0) f(x^*) dx^*,$$

where f is the $\mathcal{N}(\mu, \sigma_x^2)$ density. The payoff is zero when $S_t e^{x^*} \leq X$, i.e. when $x^* \leq k$ and $k = \ln\left(\frac{X}{S_t}\right)$, so the integral starts at k :

$$c_t = e^{-r_f T} \int_k^{\infty} (S_t e^{x^*} - X) f(x^*) dx^* = e^{-r_f T} \left[S_t \int_k^{\infty} e^{x^*} f(x^*) dx^* - X \int_k^{\infty} f(x^*) dx^* \right].$$

Standardize x^* via $z = \frac{x^* - \mu}{\sigma_x}$ and $x^* = \mu + \sigma_x z$, and let φ and Φ be the standard normal pdf and cdf. Then follow the following computations:

$$\begin{aligned} \int_k^{\infty} f(x^*) dx^* &= \Phi(d_2), \quad d_2 = \frac{\ln(S_t/X) + (r_f - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \\ \int_k^{\infty} e^{x^*} f(x^*) dx^* &= e^{\mu + \frac{1}{2}\sigma_x^2} \Phi(d_1) = e^{r_f T} \Phi(d_1), \\ d_1 &= \frac{\ln(S_t/X) + (r_f + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}. \end{aligned}$$

Substitute into the pricing integral:

$$c_t = e^{-r_f T} \left[S_t e^{r_f T} \Phi(d_1) - X \Phi(d_2) \right] = S_t \Phi(d_1) - X e^{-r_f T} \Phi(d_2).$$

Thus the Black-Scholes call price and d_1, d_2 are

$$c_t = S_t \Phi(d_1) - X e^{-r_f T} \Phi(d_2)$$

$$d_1 = \frac{\ln(S_t/X) + (r_f + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

Question 4: Black-Scholes delta

Consider a European call (no dividends) with price $c_t(S_t) = S_t \Phi(d_1) - K e^{-r_f T} \Phi(d_2)$, where:

$$d_1 = \frac{\ln(S_t/K) + (r_f + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

4.1 Delta of a call

Delta is $\delta = \frac{\partial c_t}{\partial S_t}$ so differentiate $\frac{\partial c_t}{\partial S_t} = \Phi(d_1) + S_t \varphi(d_1) \frac{\partial d_1}{\partial S_t} - K e^{-r_f T} \varphi(d_2) \frac{\partial d_2}{\partial S_t}$. using $\frac{\partial d_1}{\partial S_t} = \frac{1}{S_t \sigma \sqrt{T}}$ and $\frac{\partial d_2}{\partial S_t} = \frac{\partial d_1}{\partial S_t}$, and the identity $S_t \varphi(d_1) = K e^{-r_f T} \varphi(d_2)$.

The terms with φ cancel and finally we are left with: $\boxed{\delta = \Phi(d_1)}$. Thus the Black-Scholes call delta equals the risk-neutral probability that the option finishes in the money.

4.2 Delta as a function of the underlying price

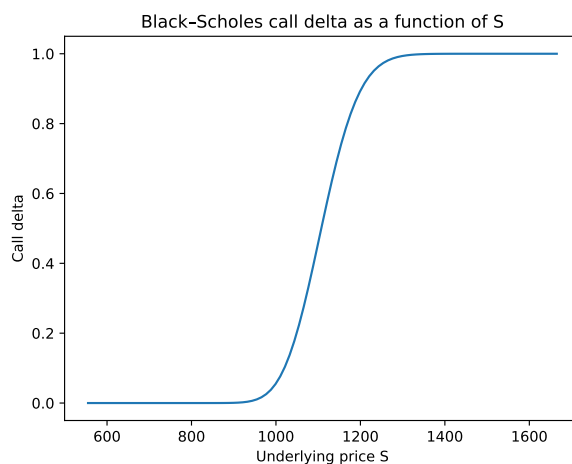


Figure 4.0.1: Historical Simulation VaR Bond

Fix (K, T, r_f, σ) and view $\delta(S_t) = \Phi\left(\frac{\ln(S_t/K) + (r_f + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right)$ as a function of S_t (or moneyness S_t/K). Delta is an increasing S-shaped curve:

$$\delta(S_t) \rightarrow 0 \text{ as } S_t \rightarrow 0, \quad \delta(S_t) \rightarrow 1 \text{ as } S_t \rightarrow \infty.$$

For the S&P 500 example in the lecture $K = 1110$, $T = \frac{43}{365}$, $r_f = 0.0025$, $\sigma = 0.1872$, the graph of $\delta(S_t)$ against S_t has a steep region near $S_t \approx K$ and flattens out deep in- or out-of-the-money. Around the steep middle region, small changes in S_t produce large changes in δ , so the linear approximation $\Delta c \approx \delta \Delta S$ is less accurate; in the flatter tails delta is nearly constant and the linear approximation is much more accurate.

Question 5: Black-Scholes implied volatility and monotonicity

5.1 Call price is increasing in volatility

For a non-dividend-paying stock the Black-Scholes call price is

$$c_{BS}(S, r_f, K, T, \sigma) = S \Phi(d_1) - K e^{-r_f T} \Phi(d_2),$$

where

$$d_1 = \frac{\ln(S/K) + (r_f + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

The derivative w.r.t. σ (vega) is $\frac{\partial c_{BS}}{\partial \sigma} = S\varphi(d_1)\sqrt{T}$. Since $S > 0$, $\sqrt{T} > 0$ and $\varphi(d_1) > 0$ for all real d_1 then it follows that $\frac{\partial c_{BS}}{\partial \sigma} > 0$ such that the Black-Scholes call price is strictly increasing in volatility.

5.2 Put price is increasing in volatility

Put-call parity (no dividends) gives $c_{BS}(S, r_f, K, T, \sigma) - p_{BS}(S, r_f, K, T, \sigma) = S - Ke^{-r_f T}$, which does not depend on σ . Differentiate w.r.t. σ :

$$\frac{\partial c_{BS}}{\partial \sigma} - \frac{\partial p_{BS}}{\partial \sigma} = 0 \quad \Rightarrow \quad \frac{\partial p_{BS}}{\partial \sigma} = \frac{\partial c_{BS}}{\partial \sigma} = S\varphi(d_1)\sqrt{T}.$$

Then $S > 0$, $\sqrt{T} > 0$, $\varphi(d_1) > 0$ such that it follows immediately that $\frac{\partial p_{BS}}{\partial \sigma} > 0$ so the Black-Scholes put price is also strictly increasing in volatility.